Problem 6.8

At a certain rate of compound interest an investment of \$1,000 will grow to \$1,500 at the end of 12 years. Determine its value at the end of 5 years.

Let the annual compound interest rate be i. Then

$$1500 = 1000(1+i)^{12}.$$

Dividing by 1000,

$$1.5 = (1+i)^{12}.$$

Taking the twelfth root,

$$1+i = (1.5)^{1/12}.$$

The value after 5 years is

$$A = 1000(1+i)^5.$$

Substituting $(1+i) = (1.5)^{1/12}$, we obtain

$$A = 1000\left((1.5)^{1/12}\right)^5 = 1000(1.5)^{5/12}.$$

Therefore,

$$\boxed{A = 1000(1.5)^{5/12} \approx 1184.04}.$$